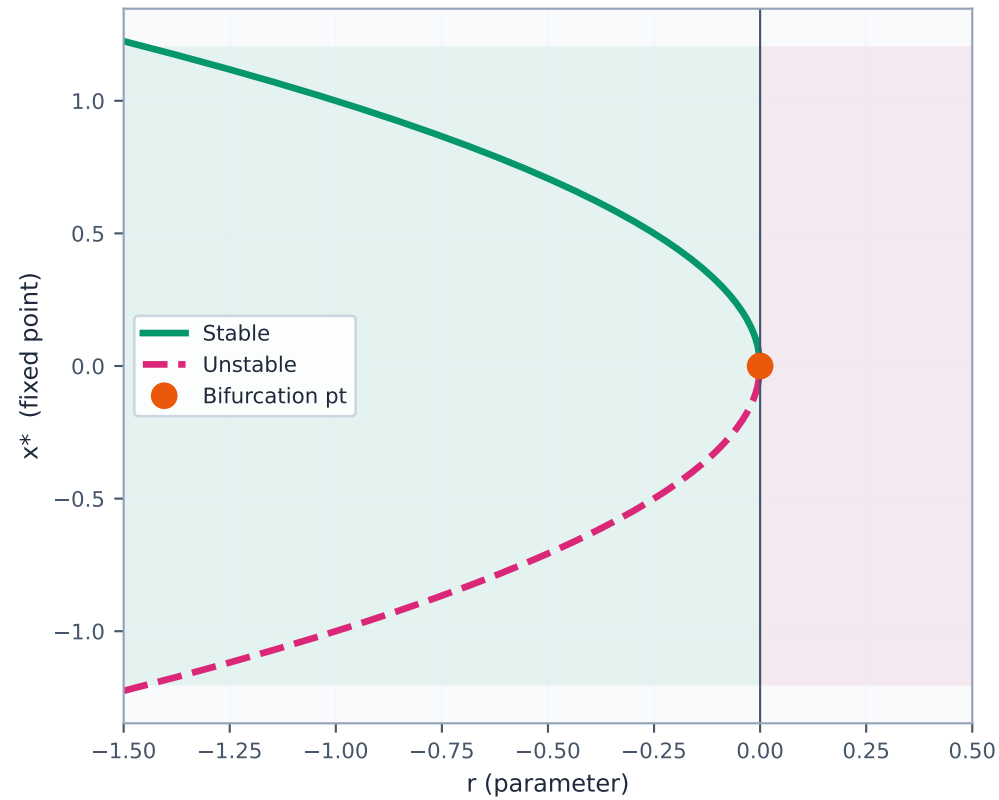
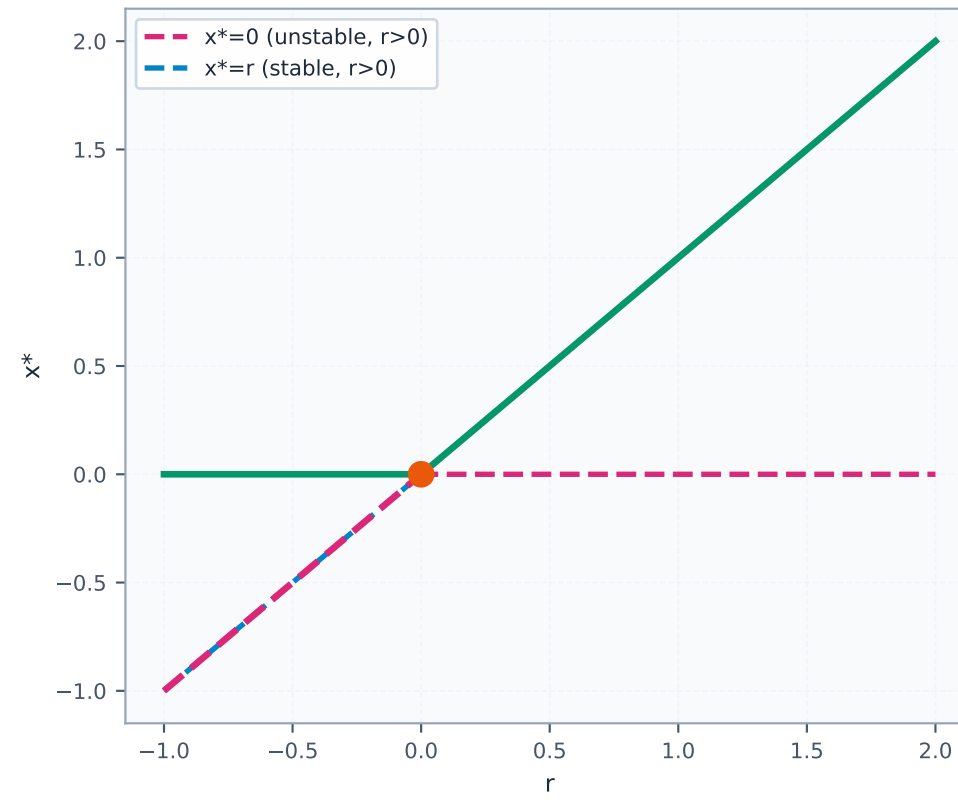


Bifurcation Diagrams in 1-D and 2-D

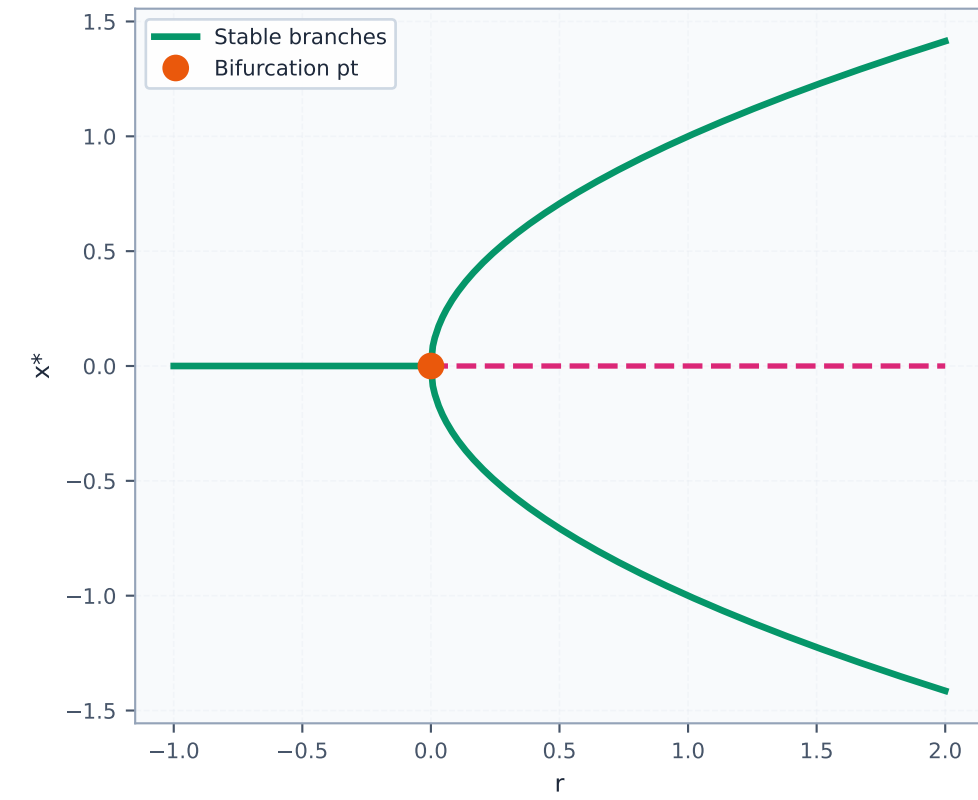
Saddle-Node $\dot{x} = r + x^2$



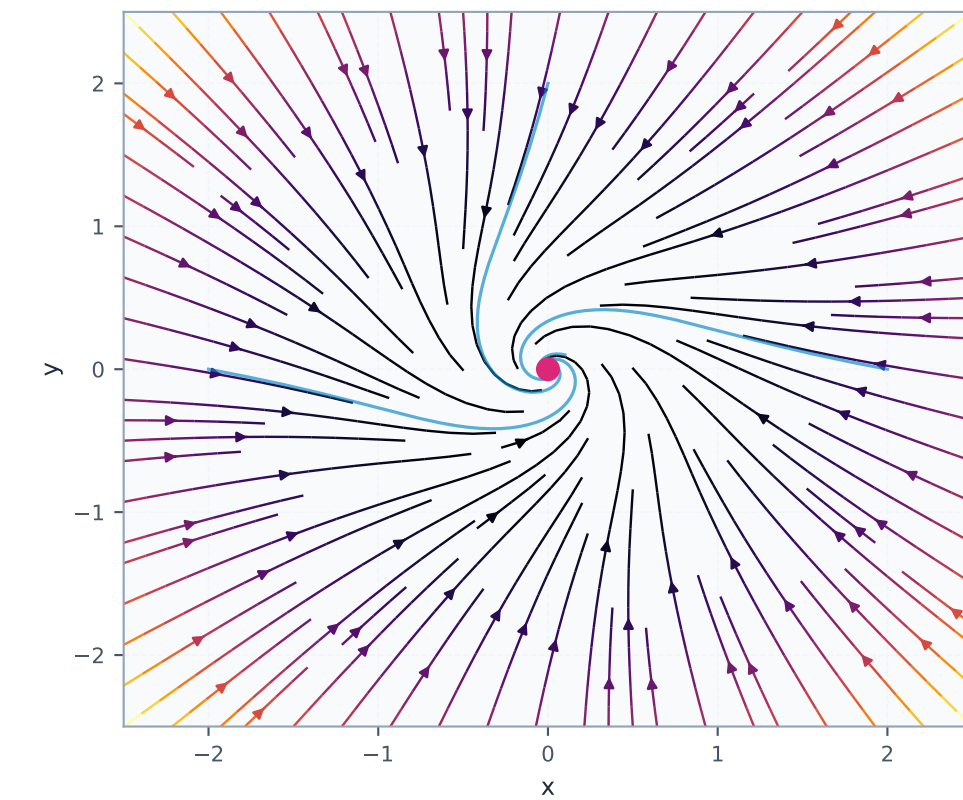
Transcritical $\dot{x} = rx - x^2$



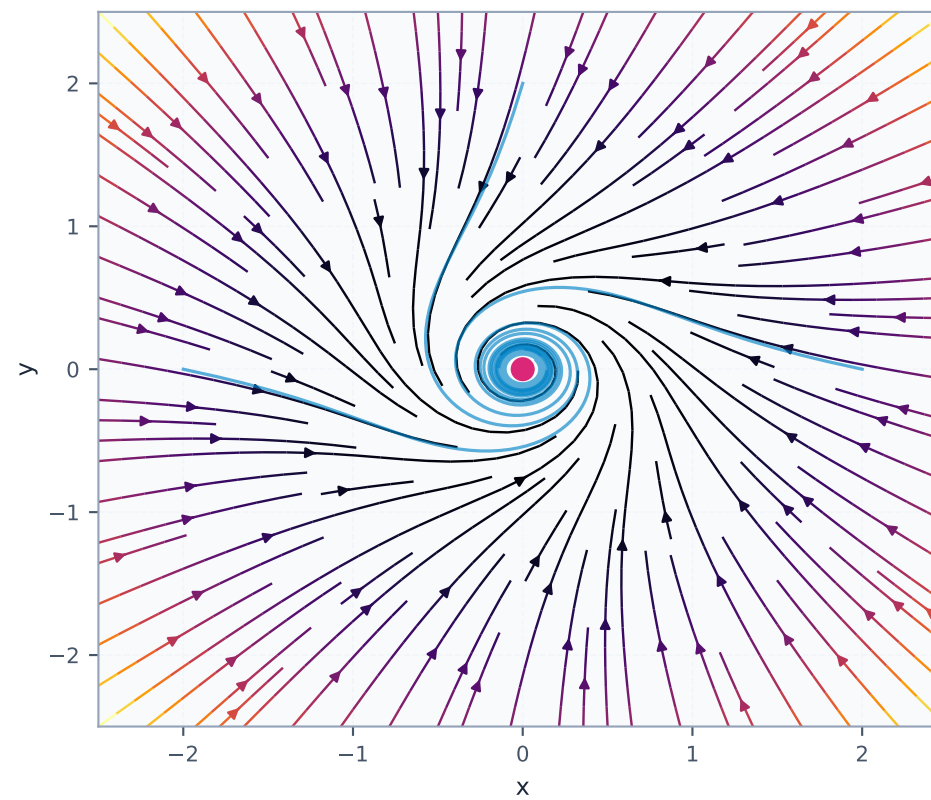
Pitchfork (supercritical) $\dot{x} = rx - x^3$



Hopf Bifurcation
 $\mu = -0.5$ (stable spiral)



Hopf Bifurcation
 $\mu = 0$ (centre)



Hopf Bifurcation
 $\mu = 1$ (limit cycle)

